Job Market Analysis & Machine Learning Project Report

1. Introduction

The rapid rise of remote work has transformed the global employment landscape. This project analyzes remote job listings to identify hiring trends, in-demand skills, salary insights, and workforce distribution patterns.

2. Problem Statement

The objective of this project was to analyze remote job market trends and build a machine learning classification model to predict job role categories based on job-related textual information.

3. Data Collection

The dataset was collected through web scraping from remote job listing platforms. The collected data included

- job titles
- companies
- locations
- skills
- salaries
- dates posted
- job links.

```
Dataset Shape:
(3909, 9)
            Category                           Job_Title             Company  \
0  remote-data-jobs  Senior Marketing Manager Paid Ads          Acebet.com
1  remote-data-jobs             DevRel Content Creator          SuperPlane
2  remote-data-jobs         Senior Fraud Risk Analyst    Braviant Holdings
3  remote-data-jobs            Senior Technical Lead   Calibrate Group AI
4  remote-data-jobs               Intern SD Engineer              Copious

            Location                                              Skills  \
0        🌍 Worldwide                                            Marketing
1          EU Europe                     Marketing, Developer, DevOps
2        Dallas, TX  Analyst, Financial, Strategy, Senior, Operatio...
3  US United States                                            Engineer
4       Maharashtra  Salesforce, Students, Founder, Support, Softwa...

             Salary Date_Posted  \
0  💰 $200k - $500k          2d
1   💰 $70k - $150k          6d
2    Not Mentioned         11h
3  💰 $160k - $200k         11h
4    Not Mentioned         19h

                                          Job_Link Description
0  https://remoteok.com/remote-jobs/remote-senior...         N/A
1  https://remoteok.com/remote-jobs/remote-devrel...         N/A
2  https://remoteok.com/remote-jobs/remote-senior...         N/A
3  https://remoteok.com/remote-jobs/remote-senior...         N/A
4  https://remoteok.com/remote-jobs/remote-intern...         N/A
```

4. Technologies & Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- TF-IDF Vectorizer
- Logistic Regression
- Naive Bayes
- Requests
- BeautifulSoup
- Selenium,
- Jupyter Notebook
- Power BI.

5. Data Cleaning & Preprocessing

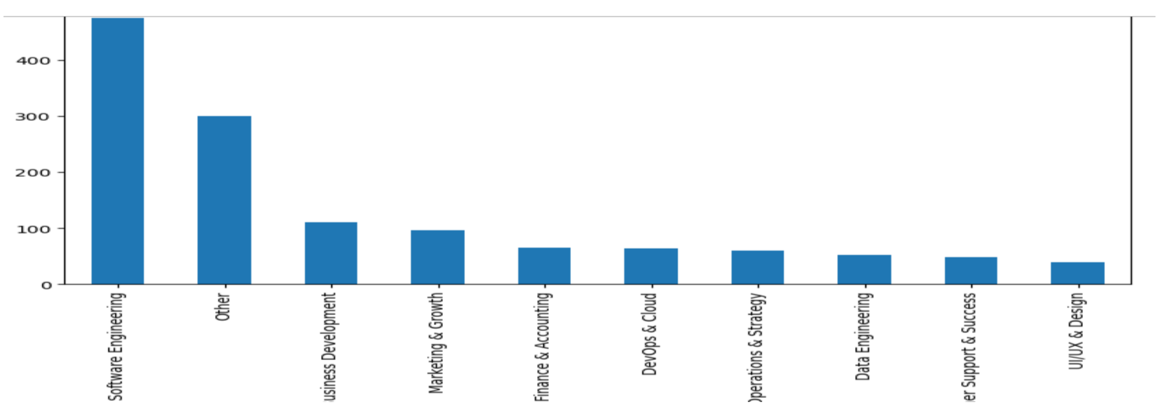
Data preprocessing involved :

- handling missing values
- removing duplicates
- standardizing locations and country names
- converting dates into datetime format
- cleaning skills data
- performing feature engineering for role categories.

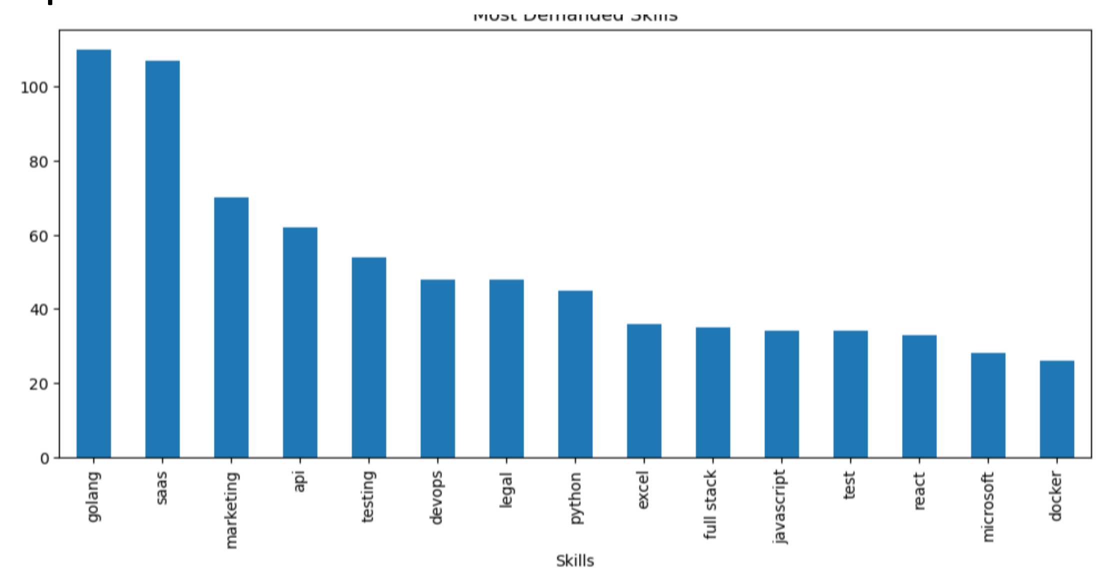
6. Exploratory Data Analysis (EDA)

EDA was performed to analyze hiring trends, top skills, country-wise hiring patterns, salary distribution, remote job trends, and company-wise recruitment insights.

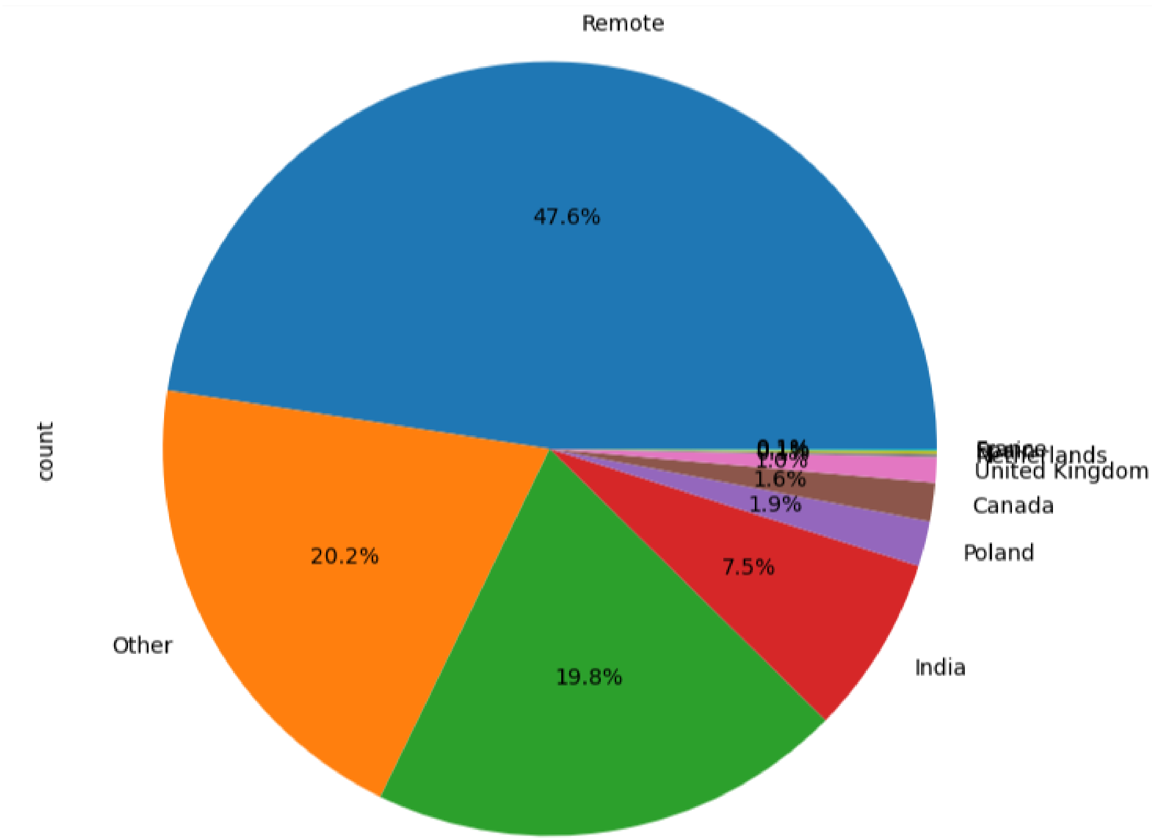
Top Job Categories :



Top Skills:



Country Distribution:



7. Key Findings & Insights

- Software Engineering roles dominated remote hiring.
- Python was among the most demanded technical skills.
- United States led global remote recruitment.
- AI, Cloud, and DevOps skills showed high demand.
- Most companies did not publicly disclose salary information.

8. Dashboard Development

An interactive Power BI dashboard was developed containing KPIs, top skills analysis, role distribution, country-wise hiring trends, salary insights, and hiring trend visualizations.

9. Machine Learning Approach

A multi-class classification model was developed to predict job role categories using Job Title, Skills, Country, and Remote status as features.

10. NLP & Feature Engineering

Textual features were combined into a single corpus and transformed into numerical vectors using TF-IDF Vectorization. Label Encoding was used for target variable encoding.

11. Model Building

The project implemented Logistic Regression and Multinomial Naive Bayes algorithms. The dataset was split into training and testing sets using an 80-20 ratio.

12. Model Evaluation

The models were evaluated using Accuracy Score, Precision, Recall, F1-Score, Classification Report, and Confusion Matrix. The confusion matrix showed strong diagonal concentration, indicating good classification performance.

```python
# Classification Report
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           1       1.00      0.91      0.95        11
           2       1.00      0.75      0.86         4
           3       0.80      0.57      0.67         7
           4       0.79      1.00      0.88        11
           5       1.00      0.17      0.29         6
           6       1.00      0.86      0.92        14
           7       1.00      0.88      0.93        16
           8       1.00      0.75      0.86         4
           9       0.81      0.91      0.86        23
          10       0.00      0.00      0.00         3
          11       1.00      1.00      1.00        15
          12       0.77      0.82      0.79        44
          13       1.00      0.33      0.50         6
          14       0.00      0.00      0.00         1
          15       0.92      1.00      0.96        23
          16       0.87      0.99      0.92        93
          17       1.00      0.67      0.80         6

    accuracy                           0.87       287
   macro avg       0.82      0.68      0.72       287
weighted avg       0.87      0.87      0.86       287
```

13. Business Impact

This project provides workforce intelligence insights for organizations, recruiters, and job seekers by identifying hiring patterns, demanded skills, and market trends.

Video Presentation :-

https://drive.google.com/file/d/15d_lbj01hSALxGobbbw4EFEoUEj-BZhV/view?usp=sharing